Data Science Foundations

Outline

- Introduction to Data Science
- Data Science Lifecycle
- Roles
- Data Science vs. AI

Acknowledgement : some slides are enhanced using AI particularly CoPilot and Gemini

What Is Data Science? 🔍

- **Definition:** collecting, analyzing, and modeling data to **generate insights** (Data Storytelling) that drive **decision-making**

📊 **Data → 💡 Insight → ✅ Decision**

🎯 **The goal:** better decisions, not pretty charts

  - Learning about the world from data using computation
  - Not just reporting: asks new questions, not just old ones

- **Interdisciplinary:** math, programming, machine learning, domain knowledge
- **Structured and unstructured data:** tables, text, images, audio, sensors data.

Difference Between Data, Information, and Insights

Level	Example
Data	Sales = $100,000 in October, $130,000 in November
Information	Sales increased by 30% from October to November
Insight	Holiday shopping behavior drives demand, creating an opportunity for seasonal promotions

A good insight is:

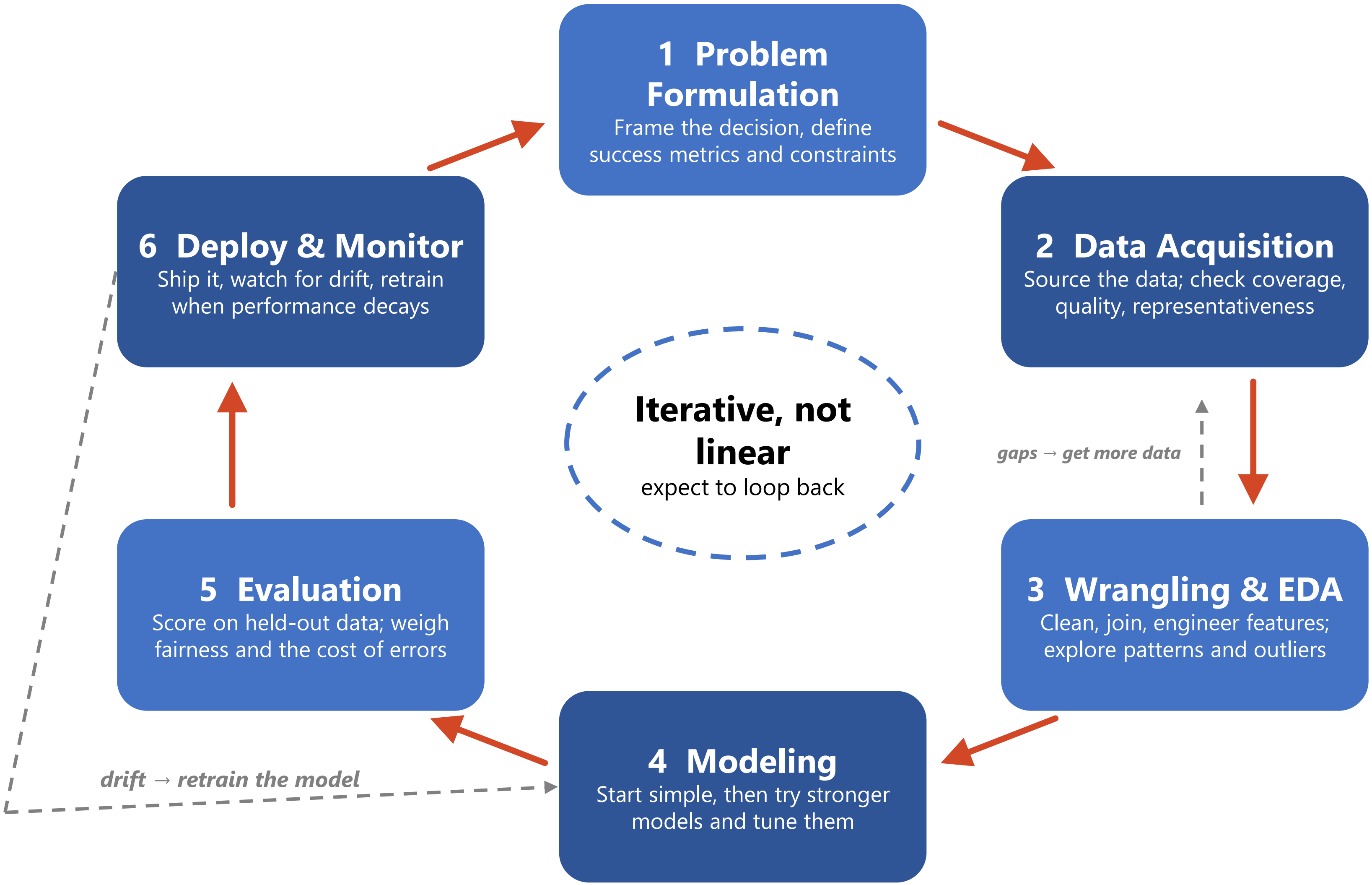
- **Relevant** to the business or research question
- **Actionable** and useful for decision-making
- **Evidence-based** and supported by data
- **Non-obvious**, revealing something not immediately apparent
- **Impactful**, leading to improved outcomes

Where Data Science Creates Value 🌍

- 🏥 **Healthcare:** imaging diagnosis, readmission risk, drug discovery
- 💳 **Finance:** fraud detection, credit scoring, risk management
- 🛒 **Retail:** demand forecasting, recommenders, dynamic pricing
- 🚚 **Logistics:** route optimization, predictive maintenance, inventory
- 🎓 **Education:** admissions analytics, early warning for at-risk students
- ⚡ **Energy:** demand forecasting, policy evaluation

Data Science Lifecycle 🔄

Data Science Lifecycle



Types of Data Analysis 📈

- 📋 **Descriptive** — "What happened?" Dashboards and KPIs.

Example: applications, enrollment, and pass rates by program each term.

- 🔎 **Diagnostic** — "Why did it happen?" Identifies the factors and root causes behind observed outcomes through drill-down and comparative analysis.

Example: Enrollment decreased by 15%. Diagnostic analysis: decline was driven by lower admissions in CE and a reduction in international student applications.

- 🔮 **Predictive** — "What is likely to happen?" Forecasting and machine learning.

Example: flagging first-year students at risk of failing a course.

- 🧭 **Prescriptive** — "What should we do?" Optimization and recommendation.

Example: which recruitment visits and scholarship offers to prioritize.

- 🧠 **Cognitive** — "Can the system reason like a human?" LLMs and NLP over unstructured data. *Example*: an AI tutor giving feedback on student essays.

⬆️ **Rising difficulty, rising value**: each level builds on the one before it.

What we will focus on in this course

- **Data Acquisition and Wrangling — getting trustworthy data**
  - Sourcing, quality, and representativeness
  - Cleaning, transformation, integration, feature engineering
- **Exploration — finding patterns in data**
  - Exploratory data analysis (EDA) and visualization
- **Inference — drawing reliable conclusions**
  - Probability, sampling, and uncertainty
  - Statistical testing, estimation, and A/B testing
  - Causal inference and experimental design
- **Prediction — modeling to make informed guesses**
  - Regression and classification
  - Clustering and dimensionality reduction

Roles

Who Does What: Roles Across the Data Science Lifecycle

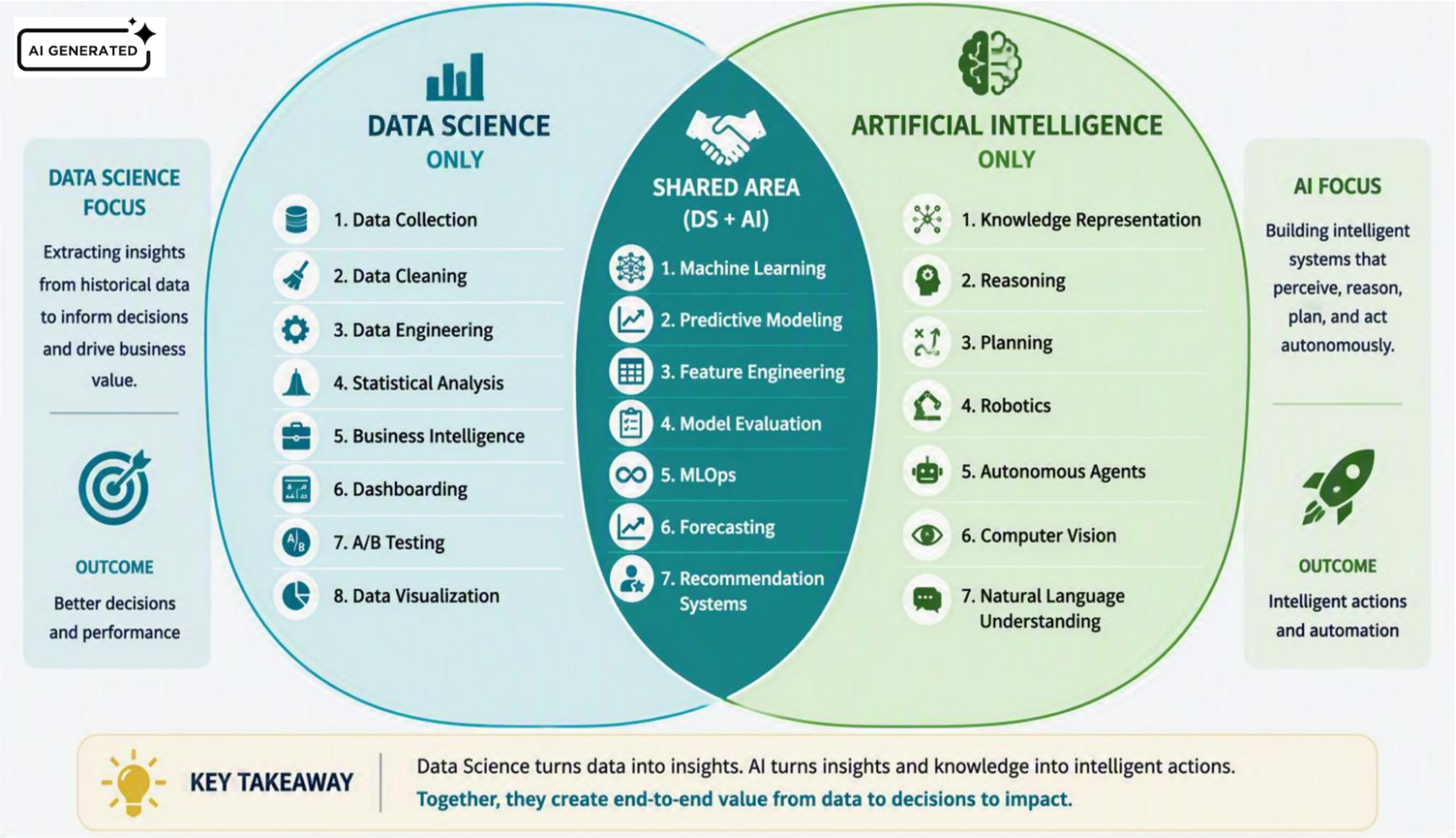
Role	The question they answer	What they deliver	Typical tools
Data Engineer *Owns stage 2, enables stage 3* **Role:** Builds and operates the jobs that land raw source data in the warehouse on schedule.	*"Can this data be trusted, and will it arrive on time?"*	Ingestion pipelines, curated tables, freshness and quality checks	SQL, Spark, Airflow, cloud data platforms
Data Analyst *Owns stage 3, informs stage 1* **Role:** Defines the metric, tracks it, and explains what is driving the movement.	*"What happened, and why do the numbers look like this?"*	Metric definitions, dashboards, and trend analyses	SQL, BI tools, spreadsheets, visualization
Data Scientist *Owns stages 1, 4 and 5, shares 3* **Role:** Frames the question, runs the experiment, and build and validates stats/ML model	*"Why did it happen, and what is likely to happen next?"*	Framed problems, experiments, validated models, evidence	Python/R, statistics, A/B testing, ML algorithms
ML Engineer *Owns stage 6, supports stage 4* **Role:** Packages a trained model as a monitored service and retrains it when performance drifts.	*"Will this model run reliably in production?"*	Deployed services, monitoring, retraining and rollback	MLOps, Docker, CI/CD, model registries
AI Engineer *Owns stage 6 for LLM products, evaluates in 5* **Role:** Builds a user-facing application on a foundation model, grounded in the organisation's own content.	*"Can we build a useful product on a foundation model?"*	LLM applications, retrieval and guardrails, evaluation suites	Prompting, RAG, agents, evaluation harnesses

The analyst's finding becomes the scientist's question, and the scientist's model becomes the ML engineer's service.
In a small team, one person wears several hats.

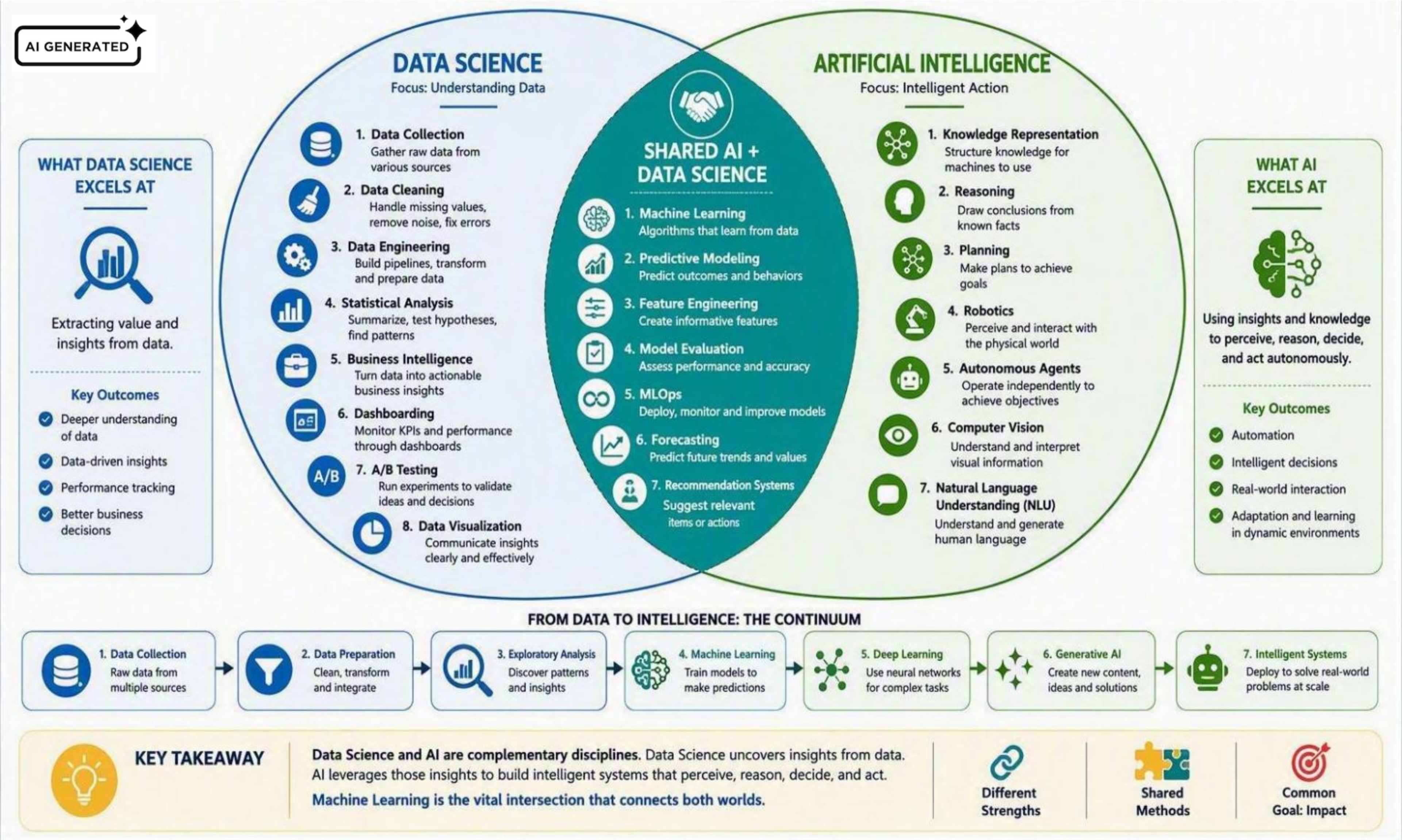
Data Science vs. AI

AI and DS: Overlaps and Unique Characteristics

Two disciplines with shared foundations and distinct strengths



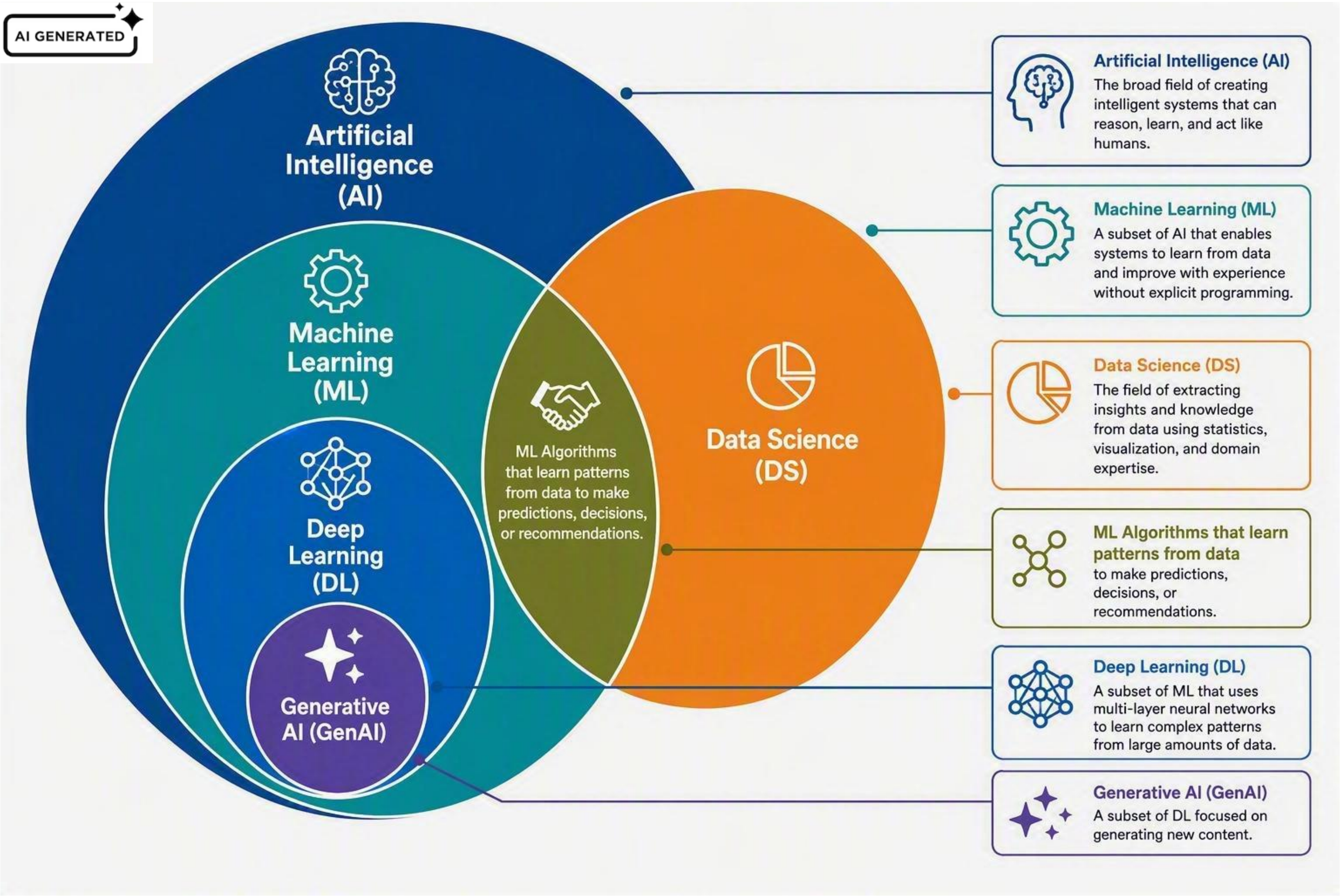
DS extracts insights from data. AI uses those insights to create intelligent systems



How DS, ML, DL, and Gen AI Relate

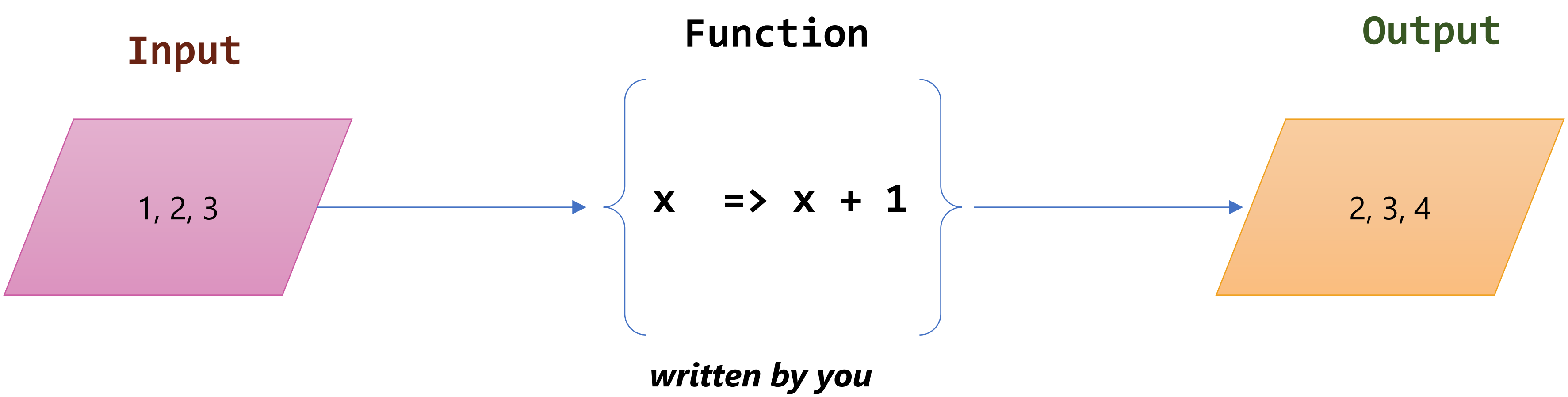
- **Data Science**: Extracts insights, predictions, and business value from data using statistics, computing, and domain knowledge.
- **AI**: Builds systems that perceive, reason, learn, and act to accomplish tasks that normally require human intelligence.
- **ML** (Intersection): Algorithms that learn patterns from data to make predictions, decisions, or recommendations.
    - **Deep Learning**: ML approach using neural networks to learn complex representations from large-scale data.
    - **Generative AI**: Models that create new content such as text, images, audio, video, and code.

Data Science vs. AI



Coding : Input + Function → Output

- You already know the mapping from input to output, so you **write the Function** yourself — the computer just applies it to every Input
- In **ML the mapping is unknown**: you supply Input and Output, and the model learns the Function

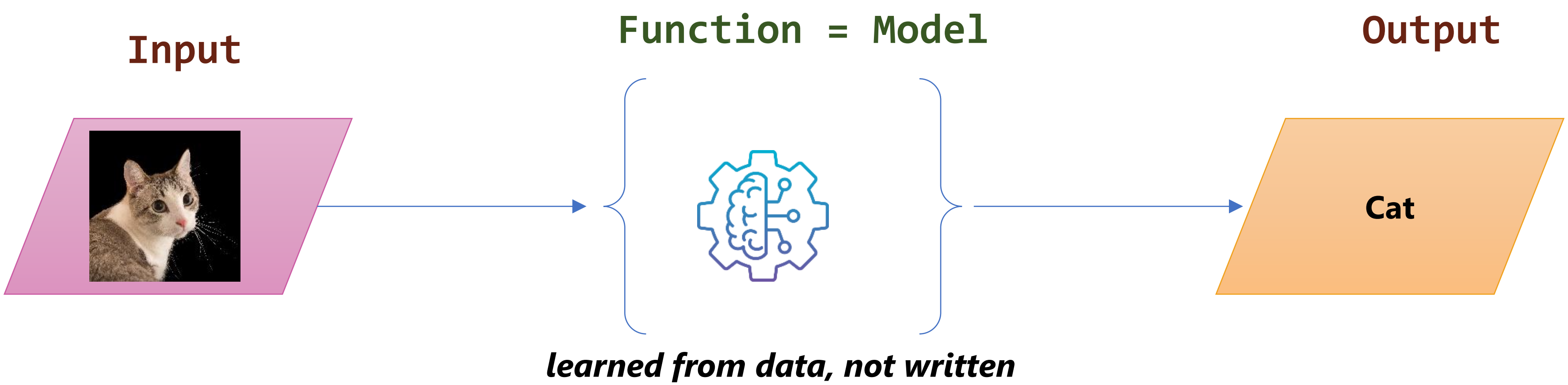


Coding: the function is known, the outputs are computed

Model building: the outputs are known, the function is learned

ML : Input + Output → Function (Model)

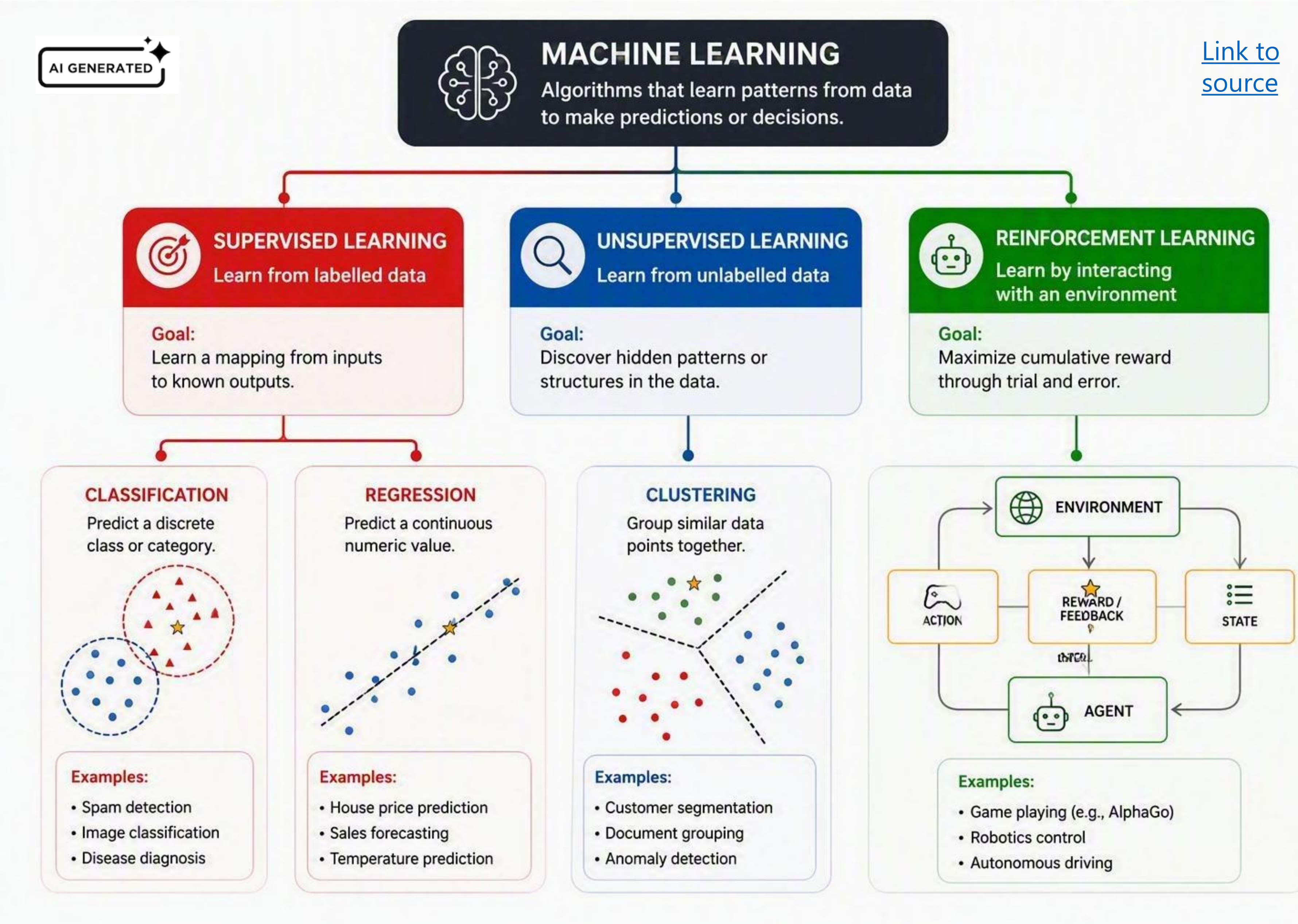
- No one can write a function for “is this a cat?” — so you collect thousands of **labelled examples**: an Input with its known Output
- Training finds the function that best **approximates** that mapping
  - The learned **function** is what we call the **model**. It can then be used to classify images it has never seen before



Training: many (image, label) pairs → the model

Prediction: a new image → the trained model → its label

Types of ML



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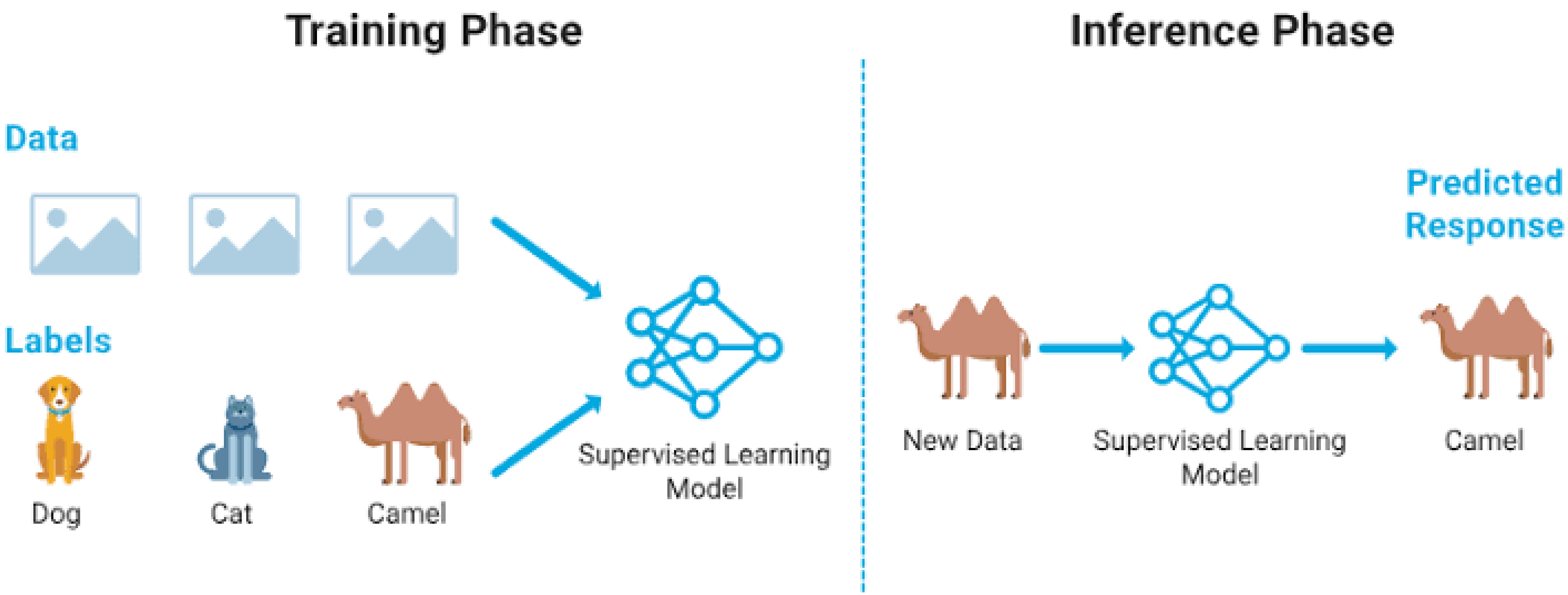
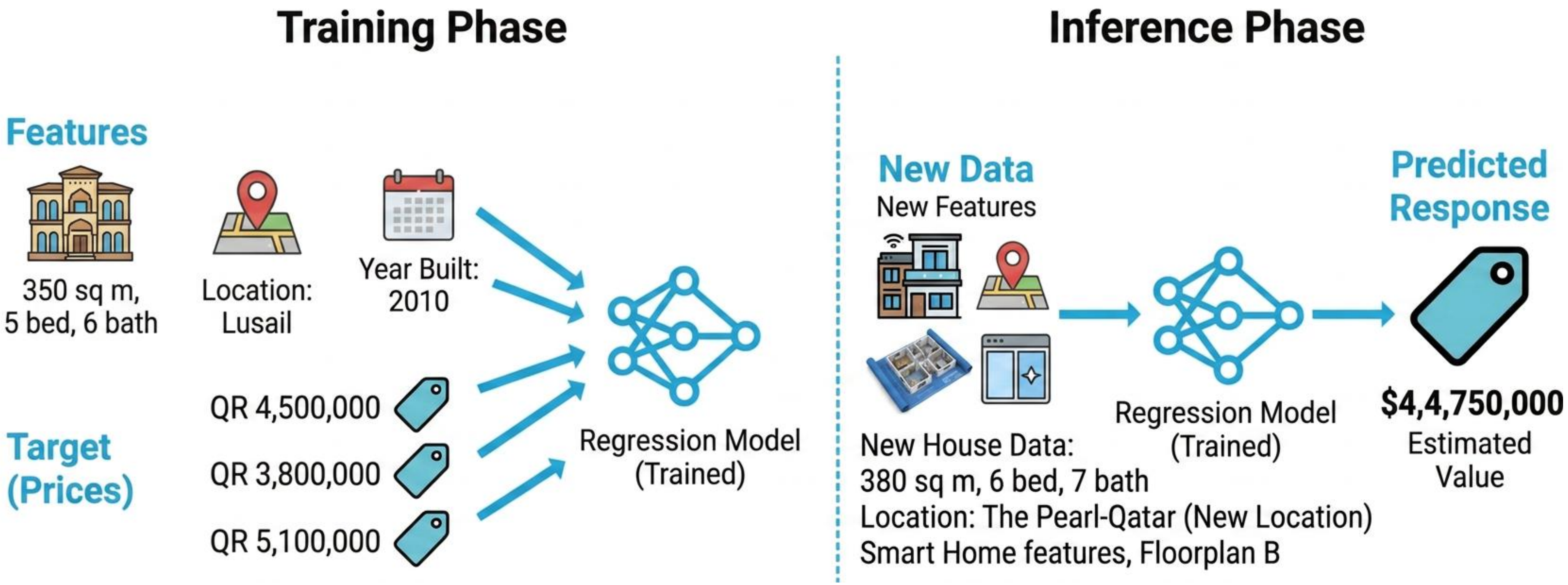
Supervised Learning – Classification Example



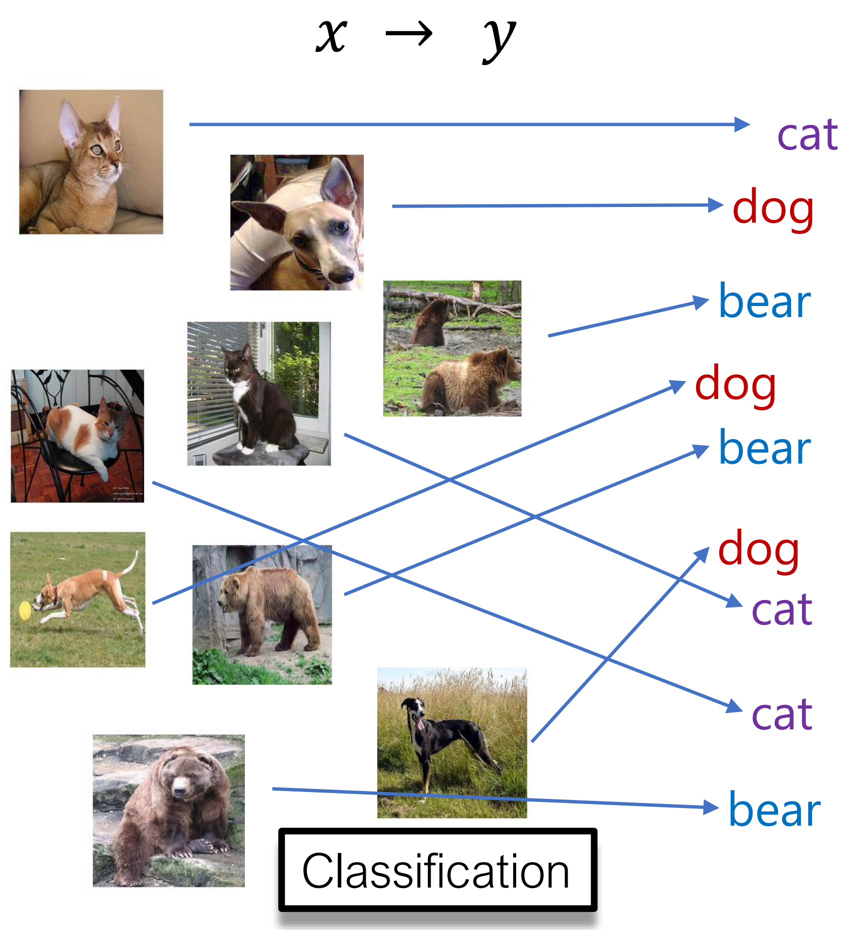
- **Inference**: apply the **trained prediction function** to a new image to predict its class

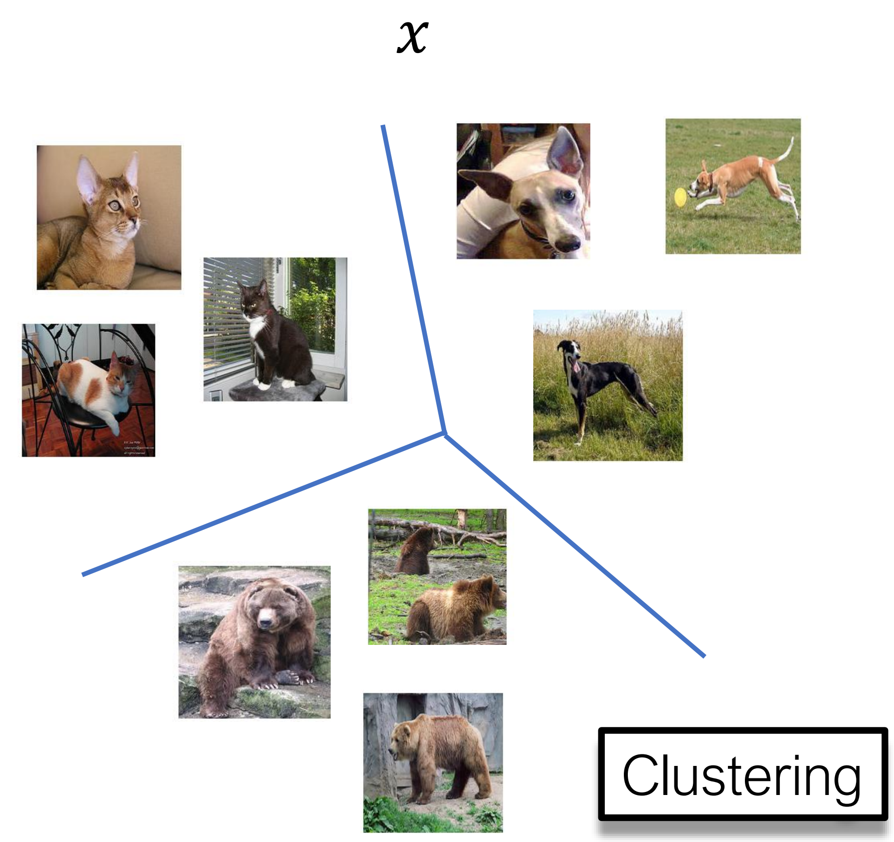
Supervised Learning – Regression Example



- **Inference:** Apply the trained **regression function** to the house's features (such as size, location, and year built) to predict its estimated value.

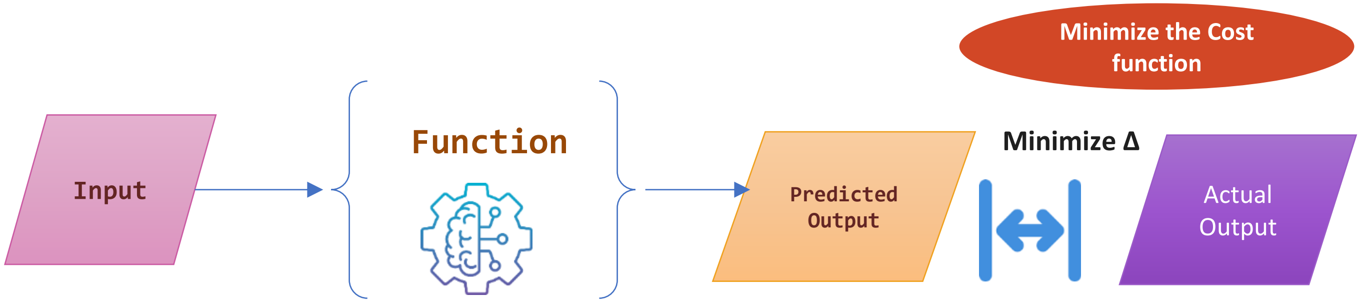
Supervised Learning vs Unsupervised Learning





ML: learn a Function that minimizes the cost

- **Guess** — start with random parameters, so the first predictions are wrong
- **Measure** — the cost function scores how far the Predicted Output sits from the Actual Output
  - also called the loss or error function
- **Adjust** — nudge the parameters to lower the cost, then repeat until it stops improving



Example: predicting house prices

- The model: **Price = w × Size + b**
  - w and b are the two numbers to learn from past sales
- **Guess** — start at w = 0, b = 0: the line is flat and the error is huge
- **Measure** — the Mean Squared Error (MSE) scores the line: the average squared gap between predicted and actual price
- **Adjust** — tilt it with w and shift it with b so the MSE drops, then repeat
- Result: **w ≈ QR 12,000 per sqm, b ≈ QR 150,000** — the MSE stops falling, so the line has settled at Price ≈ 12,000 × Size + 150,000

Blue dots = past sales | Red line = the model

w (slope) — each extra sqm adds about QR 12,000

b (intercept) — the baseline QR 150k before size counts

Cost function — Mean Squared Error: MSE(w, b) = (1/n) Σ (Actual price − Predicted price)^2

Training is the search for the w and b that make the MSE smallest.

Resources

- **Foundations & data wrangling** — Python Data Science Handbook
- **Exploratory analysis & visualization** — Fundamentals of Data Visualization
- **Probability, sampling & uncertainty** — Seeing Theory (Brown University)
- **Machine learning, hands-on** — Google Machine Learning Crash Course
- **Clustering & dimensionality reduction** — scikit-learn User Guide
- **Statistical testing & A/B experiments** — Microsoft Experimentation Platform
- **Causal inference & experimental design** — Causal Inference